Was magnitude estimation necessary?

A secondary-data reassessment of Bard, Robertson, and Sorace (1996)

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Abstract

This paper reassesses Bard, Robertson, and Sorace's case for magnitude estimation in linguistic acceptability research. It separates two claims that are often run together: acceptability judgments are graded, measurable, and methodologically serious; magnitude estimation has a practical advantage over simpler formal judgment tasks. The first claim is the stronger legacy. The second requires later method-comparison evidence.

The paper reports a constrained secondary analysis of public Sprouse method-comparison data, within an approved reuse boundary and guided by an analysis charter fixed before outcome analysis. The reanalysis finds strong cross-method convergence and bounded response-function curvature, but no systematic practical resolution advantage for magnitude estimation in these data. Langsford et al. enter as article-level evidence about reliability and response-style risks, while computational benchmarks remain afterlife evidence rather than evidence about magnitude estimation itself.

Keywords: magnitude estimation; acceptability judgments; grammaticality judgments; secondary data; experimental syntax

1 The problem

Bard, Robertson, and Sorace made a methodological claim that was larger than a single elicitation format. Acceptability judgments, they argued, could be treated as graded measurements rather than as informal binary verdicts (Bard et al., 1996). That move helped make experimental syntax a measurement enterprise: judgments could be collected from naive participants, scaled, compared across groups, and subjected to ordinary statistical tests.

The harder question is whether magnitude estimation was necessary for that advance. Magnitude estimation promised a way around the scale restrictions of traditional judgment tasks, giving participants an open response scale and preserving fine distinctions. But later acceptability work did not settle into one privileged elicitation method. Likert ratings, forced choice, yes/no tasks, and Thurstonian models all became part of the measurement toolkit (Langsford et al., 2018; Sprouse & Almeida, 2017; Sprouse et al., 2013).

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This paper separates two claims that are often allowed to travel together. One claim is strong and still plausible: acceptability is graded, measurable, and methodologically serious. The other is weaker: magnitude estimation has a practical measurement advantage over simpler formal tasks. The first claim can survive even if the second does not.

The empirical half of this is not untouched. The methodology literature has already compared elicitation formats and largely deflated magnitude estimation's special status. Schütze (2016) reports that most or all of its advertised advantages over Likert tasks have since been refuted, pointing to Weskott and Fanselow (2011) and Sprouse et al. (2013); later work weighed the formats on convergence (Sprouse et al., 2013), statistical power (Sprouse & Almeida, 2017), and reliability and response style (Langsford et al., 2018).

So the bare verdict that magnitude estimation carries no special advantage is not what this paper adds. The contribution is to separate that verdict from the durable measurement claim, to relocate acceptability within a projectibility account, and to route each later dataset to an analysis-chartered estimand, so that any approved reanalysis of public comparison data cannot quietly select whichever scale looks best.

The reassessment is therefore not a rejection of Bard et al.'s programme. It is a test of what should be credited to magnitude estimation itself, and what should instead be credited to the broader formalization of acceptability judgment methodology.

2 The Bard target

Reassessing Bard et al. means first stating, from the article rather than its reputation, what they actually claimed. The abstract makes four claims for magnitude estimation: that it can "solve the measurement scale problems which plague conventional techniques", "provide data which make fine distinctions robustly enough to yield statistically significant results of linguistic interest", be "usable in a consistent way by linguistically naive speaker-hearers", and "allow replication across groups of subjects" (Bard et al., 1996, p. 32).

The core of the case is an argument about scale type, not about reliability. The trouble with conventional annotation, they argue, is "the use of the wrong kind of measurement scale" (Bard et al., 1996, p. 32): the asterisk system, like any fixed-point scale, "predetermines the number of distinctions subjects may use" and so censors finer ones (Bard et al., 1996, p. 35). Magnitude estimation is the proposed remedy because it "does not restrict the number of values", giving subjects "complete freedom about which of the infinite set of numbers to use" and yielding interval and ratio data (Bard et al., 1996, p. 41).

This sharpens what is, and is not, Bard et al.'s claim. They do make a method-superiority claim, but a specific one: an open, ratio-yielding response recovers gradient distinctions that fixed-point scales lose by construction. That is a claim about resolution and scale-boundary censoring, and it is testable. The broader reading sometimes met in reception, that magnitude estimation is the privileged formal method in general, goes beyond the article; Bard et al. compare magnitude estimation with informal annotation and fixed-point scales, not with the later inventory of Likert, forced-choice, yes/no, and Thurstonian tasks.

Bard et al. are also more tentative than the reception. They grant that the advantages "are garnered at considerable empirical cost", and they leave open which psychological model underlies acceptability, since "the data currently do not decide between them" (Bard et al., 1996, p. 65). The fair target

Source	Evidence level	Use in this paper	Claim it cannot support
Bard et al. 1996	Published article	Historical target and scale-resolution claim	Direct replication without participant rows
Sprouse 2013/2017	Public row-level comparison data	Constrained secondary method comparison within approved reuse boundary	Raw-data redistribution or drift beyond the approved methodological scope
Langsford et al. 2018	Published article and supplement	Reliability and response-style synthesis	New participant-level reliability model without raw rows
CoLA, BLiMP, SAD	Benchmark labels and documentation	Afterlife of formal acceptability measurement	Evidence about ME versus other human judgment methods

Table 1: Evidential routing before analysis. ME = magnitude estimation; SAD = Syntactic Acceptability Dataset.

for reassessment is the resolution claim, not a gold-standard claim the authors did not make.

The distinction matters for design. If the live claim is that an open ratio scale recovers distinctions fixed scales censor, then the later method comparisons have to test resolution and scale-boundary effects alongside convergence and reliability. And because no public Bard participant-level rows have been found, this paper cannot replicate the original experiment directly; the reassessment is a conceptual replication and secondary-data test of whether the resolution claim retains independent evidential force.

3 Secondary-data design

The secondary-data design treats the project as a measurement reassessment of Bard, Robertson, and Sorace (Bard et al., 1996). Four kinds of source should stay separate (distinct from the evidential channels of Section 6; these are sources routed to estimands, not channels into a profile). Published summaries from Bard et al. can source the historical claim, but not a direct replication unless their participant-level data is found.

Later judgment studies can support secondary analysis when they compare elicitation methods on shared or comparable items and provide the rows needed for the intended estimand. Computational benchmarks can show the afterlife of acceptability labels, but they do not bear on magnitude estimation versus Likert ratings. Construct-level discussion can separate acceptability from grammaticality, naturalness, standardness, and correction.

The approved reuse boundary resolves the Sprouse side of the genre gate. Public row-level data from the Sprouse comparison studies can now support a constrained secondary analysis of method performance. The Langsford et al. reliability study is essential evidence for the argument, but unless its raw participant responses are obtained it can only constrain the paper at the article-summary level. The paper should therefore not promise a full response-style or test-retest model across the whole literature.

The Sprouse files also require representation choices before modelling. For Sprouse et al. (2013), the primary forced-choice comparison should use the pair-level sign-test file, because forced choice is a pairwise task and that file already records judgments against expected patterns. The logistic file preserves item-level binary rows, changing the unit of analysis, so it belongs as a named sensitivity specification.

For Sprouse and Almeida (2017), yes/no rows with item IDs `B`, `G`, and `M` should be excluded from item-level comparisons. They are absent from the materials workbook, occur in repeated early order positions for every subject, and do not align with the magnitude-estimation, Likert, or forced-choice item universe. This exclusion is a structural inclusion rule, not an outcome-based trimming decision.

Before any modelling, each source should be assigned to an estimand it can actually answer. The primary estimand is the item-level acceptability signal recoverable across methods. Secondary estimands are method-specific noise or bias and decision agreement for theoretically relevant contrasts.

Participant response style and method-by-item instability are admissible only when the dataset contains the participant-level rows and repeated structure needed to estimate them. This ordering keeps the analysis from selecting whichever scale looks best after the fact, the kind of forking path Gelman and Loken warn about (Gelman & Loken, 2013, 2014).

The substantive comparison is a measurement null, but a careful one. Magnitude estimation, Likert ratings, forced choice, and Thurstonian choice are modelled as readings of a common item-level acceptability signal through method-specific response functions, not as the same signal plus method noise: the bounded methods can compress the signal at the extremes, while the open ratio scale need not. The single-dimension assumption should be checked before any method-specific term is read, by comparing a one-factor model against correlated-two-factor and bifactor alternatives and checking method residuals for item-level structure where the data support that comparison.

This ordering matters because the advantage Bard et al. actually claimed is about resolution, recovering distinctions that fixed scales censor (Bard et al., 1996, p. 35). A plain common-signal-plus-noise null would book that resolution as magnitude-estimation noise and let the method only lose; the response functions and the dimensionality check let a resolution advantage register instead. If one factor suffices, magnitude estimation still has to earn its term; if it does not, the analysis turns to where the methods diverge. Scrambled labels, method-label permutations, and simulations from the response-function model remain negative controls for the pipeline, not the substantive comparison.

4 Scale and reliability

The scale question should be phrased as practical measurement advantage over the response-function measurement null set out in the secondary-data design. Bard et al.'s case for magnitude estimation included scale resolution, fine-grained distinctions, naive-participant use, and replication across groups (Bard et al., 1996, p. 32). The safer target is that source-grounded claim, together with the stronger reading of magnitude estimation that took hold in parts of the later methodology literature.

Later method-comparison data give the paper a sharper question: does magnitude estimation recover item differences, participant agreement, or theoretically relevant decisions that simpler tasks lose, and in particular does it recover graded distinctions that bounded scales compress at the extremes?

Sprouse et al. (2013) compare formal judgment tasks with informal published judgments, giving evidence about convergence between magnitude estimation, Likert ratings, and forced choice. The public Sprouse rows can support a bounded reanalysis of item-level agreement and decision consequences within the approved reuse boundary. Langsford et al. (2018) add a reliability and response-bias target by comparing Likert, magnitude estimation, forced choice, and Thurstonian models. Without their raw rows, Langsford can support the synthesis of published reliability findings but not a new participant-level response-style or test-retest model.

The reliability analysis inherits the unit-of-analysis charter fixed in the secondary-data design rather than reopening it. The same primary and sensitivity representations carry over, so reliability claims stay traceable to fixed units of analysis instead of to whichever representation later proves most convenient.

The primary comparison weighs methods on resolution, prediction, decision consequences, and convergence, prioritising consequences for the linguistic contrast. Resolution is the criterion Bard's claim lives on: whether magnitude estimation recovers graded item differences that bounded scales compress at ceiling or floor, or a measurement dimension that survives the dimensionality check. Reliability is a distinct property, not a proxy for it; Langsford et al. (2018) report magnitude estimation as less reliable, which does not by itself settle resolution.

For this paper, a practical magnitude-estimation advantage would have to change an inferential decision. A summary-statistic improvement alone would not count. It would count if magnitude estimation recovered a good-vs-bad contrast that bounded methods missed, showed its largest extra spread exactly where bounded scales compressed the response, carried a stable method-specific dimension across scoring choices, or reduced sign-error or exaggeration risk for a named contrast under a design analysis. A larger correlation, by itself, would not settle the question.

The Sprouse reanalysis gives a narrow answer. Magnitude estimation and the bounded methods share a strong item-level signal. Raw Likert means also show the expected bounded response-function curvature relative to magnitude estimation. That curvature is the pattern Bard's scale argument makes visible, but the endpoint and pair-level diagnostics do not show a systematic practical distinction preserved only by magnitude estimation.

The dimensionality result is deliberately modest. It is a first-pass aggregate check of the item and contrast matrices, used to ask whether a clear multidimensional rescue for magnitude estimation appears in the available Sprouse summaries. It does not replace a full participant-level factor model, which would require a broader row-level evidence base than this paper has.

Table 2 therefore separates three facts. First, the aggregate matrices provide no obvious multidimensional rescue for magnitude estimation. Second, the Likert scale behaves like a bounded response function, so a fair null should allow compression. Third, the available Sprouse contrasts do not show the further pattern that would make magnitude estimation practically superior: endpoint regions are not where magnitude estimation spreads out most, and good-vs-bad pairs ranked highly by magnitude estimation are also ranked highly by bounded methods.

These aggregate practical diagnostics are weaker than posterior certainty statements or formal equivalence tests. They show that the available contrasts lack the predicted ME-only pattern; they leave open participant-level method factors, response-style dimensions, and locally dependent item effects.

Type S and Type M errors enter only as properties of estimating a named contrast at a given sample size, not as properties a format carries. The relevant question is whether choosing magnitude estimation rather than another method changes the sign-error probability or the exaggeration ratio for a specific contrast under a design analysis (Gelman & Carlin, 2014). Sensitivity checks should vary

Diagnostic	Sprouse et al. 2013	Sprouse and Almeida 2017
ME–Likert convergence	Condition $r = .986$; pair $r = .963$	Item $r = .925$; pair $r = .944$
First-pass dimensionality check	Aggregate pair matrix: first component 89.4%; second eigenvalue below parallel-analysis threshold	Aggregate item and pair matrices: first components 80.6% and 85.9%; second eigenvalues below thresholds
Bounded response function	Bounded logistic improves over linear; endpoint ME SDs .142 and .175 versus middle .313	Bounded logistic improves over linear; endpoint ME SDs .249 and .321 versus middle .349
Pair robustness	Bounded predictors $R^2 = .926$; no ME top-quartile pair falls in the bounded bottom half	Bounded predictors $R^2 = .901$; no ME top-quartile pair falls in the bounded bottom half

Table 2: Sprouse diagnostics for Bard's resolution claim. ME = magnitude estimation. The dimensionality row is an aggregate first-pass check, and the response-function rows use magnitude estimation as an unbounded diagnostic axis rather than the true latent acceptability scale.

transformations and exclusion rules as named specifications rather than as invisible post hoc choices (Steegen et al., 2016).

5 The contemporary afterlife of acceptability

Contemporary benchmark resources show what survived from the formal acceptability programme, but it is worth being exact about what survived. CoLA turns examples from linguistics publications into binary labels for model evaluation (Warstadt et al., 2019); BLiMP turns expert-designed grammatical contrasts into large sets of minimal pairs (Warstadt et al., 2020). What they inherit is the portability of contrasts, the idea that acceptability-related distinctions can be made portable, scored, and compared across systems. What they inherit it through is the expert-asterisk channel, not Bard et al.'s naive-participant graded measurement.

They also change the evidential object. CoLA's labels come from published linguistic examples rather than from a new sample of naive participant responses. BLiMP's contrasts are generated from expert-crafted grammars and then used to test whether a model prefers the acceptable sentence in a pair. Those are useful evaluation resources, but they are not method-comparison data for magnitude estimation, Likert ratings, forced choice, or yes/no judgments.

The Syntactic Acceptability Dataset is the most useful bridge case because it keeps grammatical status and crowd acceptability status apart (Juzek, 2024). That separation is exactly the level discipline this paper needs. So the split is clean: the portability of contrasts survived into CoLA and BLiMP through the expert channel, while Bard's graded naive-participant measurement has a closer analogue in crowdsourced resources like the Syntactic Acceptability Dataset. Benchmark labels show the afterlife of one channel; participant-response datasets are still needed to compare human elicitation methods.

6 Grammaticality and acceptability

The reassessment also needs a disciplined vocabulary. The background view adopted here is that grammar is a system of projectibility profiles (Goodman, 1955): relatively stable patterns that license inferences from forms, meanings, contexts, speakers, and judgments to uses a community or register treats as available. Profiles are projectible for a purpose: here, predicting how a construction behaves across speakers, registers, and tasks well enough to support theoretical and applied syntactic claims. A grammaticality claim is therefore not a report of one response. It states what a form projects to: interpretation, production, correction, expert classification, register fit, cross-speaker stability, and related consequences.

Three of these are observation channels, not the target. Participant responses under an acceptability prompt are observations of judgments under a task. Expert asterisks and published example labels are classifications made within a theoretical or editorial practice. Benchmark labels are dataset-construction decisions. The profile itself is the fourth thing: the structure of inferences a form licenses, what its acceptability, production, correction, expert classification, register fit, and cross-speaker stability jointly project to. Two channels are evidence about the same profile when they bear on the same form's projection relations.

So the profile is not the average of the channels, and channel disagreement is not noise to smooth away. When a sentence is rated low by naive participants, left unstarred in the literature, and attested in a register corpus, the verdict about its profile is fixed by which inferences actually hold, tested against further data, not by a vote across channels.

A channel that consistently fails to predict the others is then evidence about that channel, not about the profile. This is what makes the profile more than an unweighted set of labels, and it is the identity condition the level discipline needs.

This discipline keeps neighbouring constructs apart. Acceptability is one evidential route into a projectibility profile, not the profile itself. Personal production, naturalness, standardness, correction behaviour, and theoretical classification project differently. A sentence can be dispreferred without being ungrammatical, correctable without being unnatural, or standard in one register while unacceptable under another task framing.

The framing earns its place in the measurement argument by predicting where a single acceptability scalar should be insufficient: items where participants rate a sentence low but the literature leaves it unstarred, or where the form is standard in one register and degraded under another task framing.

Those are the items where the dimensionality check in the secondary-data design should fail to support a single scalar, and where magnitude estimation's open scale, if it carries a second dimension, should diverge from the bounded tasks. The projectibility account therefore makes a testable prediction about the method comparison rather than a distinction alone.

Bard et al. helped make one part of this space measurable: graded acceptability judgments from participants (Bard et al., 1996). That is a major contribution to the measurement of one evidential channel. It does not by itself show that an acceptability scale captures every inference licensed by a grammar. Later benchmarks sometimes repackage acceptability-related labels as grammaticality targets (Warstadt et al., 2019, 2020). The Syntactic Acceptability Dataset is valuable here because it explicitly separates literature-derived grammatical status from crowd acceptability status (Juzek, 2024).

That boundary also limits the no-new-data paper. Existing datasets can compare the stability of elicitation methods and clarify how acceptability labels travel. They cannot settle a full projectibility profile for a construction: production, correction, register, interpretation, and community licensing would require data designed for those inferences.

7 Conclusion

The balanced update is straightforward. Bard et al. were right to make acceptability measurable and methodologically serious. Their documented case for magnitude estimation concerned scale resolution, fine distinctions, naive-participant use, and replication across groups (Bard et al., 1996, p. 32). That claim deserves credit as part of the history of experimental syntax.

The method-specific conclusion is more limited, and it has to be tested fairly. Magnitude estimation is compared against a measurement null in which later formal tasks read a common item-level acceptability signal through method-specific response functions, and in which the single-dimension assumption is itself tested before any method-specific term is read. The question is then whether magnitude estimation earns its keep on resolution, the distinctions Bard argued fixed scales censor, or on prediction, decision stability, or construct interpretation, rather than whether it survives a null that books resolution as noise.

This paper approximates that design with the available Sprouse evidence: aggregate cross-method item and contrast matrices, response-function diagnostics, and pair-level robustness checks. That scope matters because a full participant-level dimensionality model across Bard, Sprouse, and Langsford would require rows that have not been found.

The Sprouse reanalysis supports that narrower update. Magnitude estimation, Likert ratings, forced choice, and yes/no judgments share a strong practical signal in the available comparison data. First-pass aggregate dimensionality checks show no obvious evidence against a single dominant dimension at that level of aggregation. Raw Likert means show bounded response-function curvature, which is exactly why the null should not treat bounded tasks as simple noisy copies of magnitude estimation.

The remaining resolution diagnostics are less favourable to the stronger method claim: endpoint regions do not show unusually large magnitude-estimation spread, and pair-level bounded predictors recover most magnitude-estimation contrast variance. Langsford et al. still enter at the article level unless their raw responses become available.

The conclusion would change under a clear ME-only pattern: endpoint regions with much larger ME spread than middle regions, high-ME contrasts that bounded methods consistently placed low, stable method-specific residual structure across scoring choices, or lower sign-error and exaggeration risk for named linguistic contrasts. Those are the patterns the present Sprouse diagnostics do not show.

The central conclusion is therefore stable. The durable legacy is the formal measurement of graded acceptability as one route into grammar's projectibility profiles. The broader privileged status sometimes read into magnitude estimation is a separate claim, and later evidence has to earn it separately.

Data and code availability

No raw participant-level data is committed to this repository. Source and dataset verification are tracked in `notes/source-verification.md`; the analysis charter and structural Sprouse inventory are tracked in `notes/analysis-charter.md` and `notes/dataset-inventory.md`. Jon Sprouse approved reuse for the limited secondary methodological analysis described in the charter after checking with Diogo Almeida.

The manuscript reports summary diagnostics rather than redistributing participant-level rows or derived participant-level tables. Scripts generate ignored local structural derivatives, readiness manifests, and descriptive item/contrast signal, dimensionality, response-function, and pair-level robustness outputs under `data/derived/`; participant-level raw files remain outside the repository.

The local reproduction recipe is documented in `analysis/README.md`. The analysis charter was versioned in Git and fixed before outcome analysis as an internal analysis record rather than an external registry entry.

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